

In-class Exercise 4

● Graded

14 Days, 10 Hours Late

Student

Jinku Jung

Total Points

4 / 4 pts

Autograder Score

4.0 / 4.0

Passed Tests

part 1 (1/1)

part 2 (1/1)

part 3 (1/1)

part 4 (1/1)

Question 2

Adjustment

0 / 0 pts

✓ + 0 pts No adjustment

+ 0 pts Point adjustment below

Autograder Results

part 1 (1/1)

part 2 (1/1)

part 3 (1/1)

part 4 (1/1)

Submitted Files

```
1
2  /*******
3  // ArrayStuff.java
4  //
5  // Repository for some static methods for manipulating arrays
6  //
7  /*******
8
9  import java.util.Arrays;
10
11  public class ArrayStuff{
12
13
14      /* returns a new array containing only the elements
15       * of stuff that were at even indices*/
16      public static int[] evenIndices(int[] stuff) {
17
18          int nAnswers = 0;
19
20          int[] temp = new int [stuff.length];
21
22          for (int i = 0; i < stuff.length; i += 2) {
23
24              temp[nAnswers] = stuff[i];
25              nAnswers++;
26
27          }
28
29          int[] answer = new int[nAnswers];
30
31          for (int i = 0; i < nAnswers; i++) {
32
33              answer[i] = temp[i];
34
35          }
36          return answer;
37      }
38
39      /* returns a new array containing only the elements
40       * of stuff that have even values*/
41      public static int[] evenValues(int[] stuff){
42
43          int evens = 0;
44
45          int[] temp = new int[stuff.length];
46
47          int i = 0;
48
49          for(int element: stuff) {
```

```

50
51     if(element % 2 == 0) {
52         temp[i] = element;
53         i++;
54         evens++;
55     }
56 }
57
58 int[] answer = new int[evens];
59
60 for(int j = 0; j < evens; j++) {
61     answer[j] = temp[j];
62 }
63
64 return answer;
65 }
66
67 /* returns a new array with the elements
68  * of stuff in reverse order*/
69 public static int[] reverse(int[] stuff){
70
71     int arraySize = 0;
72
73     arraySize = stuff.length;
74
75     int[] answer = new int[arraySize];
76
77     int i = (arraySize - 1);
78
79     while(i > -1) {
80
81         for(int j = 0; j < arraySize; j++) {
82             answer[j] = stuff[i];
83             i--;
84         }
85
86     }
87
88     return answer;
89 }
90
91 /* returns a new array of length two that
92  * has the last element of stuff first and
93  * the first element of stuff last */
94 public static int[] lastWillBeFirst(int[] stuff) {
95
96     final int arraySize = 2;
97
98     int[] answer = new int[arraySize];
99
100     answer[0] = stuff[stuff.length - 1];
101     answer[arraySize - 1] = stuff[0];

```

```
102  
103     return answer;  
104 }  
105 }
```

▼ **_MACOSX/._ArrayStuff.java**

 Download

1 Binary file hidden. You can download it using the button above.

```
1
2  /*******
3  // DataTest.java
4  //
5  // Test Class for ArrayStuff
6  //
7  /*******
8
9  public class ArrayTest{
10
11      public static void main(String[] args){
12
13          // make a random test array
14          int[] nums=new int[6];
15          for(int i=0;i<nums.length;i++){
16              nums[i]= (int) (Math.random()*101);
17          }
18
19
20          // make string versions of the test array
21          String nums_str="";
22          for(int element:nums)
23              nums_str+=element+" ";
24
25
26          // make string versions of the results
27          String nums_ind="";
28          for(int element:ArrayStuff.evenIndices(nums))
29              nums_ind+=element+" ";
30
31          String nums_val="";
32          for(int element:ArrayStuff.evenValues(nums))
33              nums_val+=element+" ";
34
35          String nums_rev="";
36          for(int element:ArrayStuff.reverse(nums))
37              nums_rev+=element+" ";
38
39          String nums_lwbf="";
40          for(int element:ArrayStuff.lastWillBeFirst(nums))
41              nums_lwbf+=element+" ";
42
43
44          // print results
45          System.out.println("starting contents: \t"+nums_str);
46          System.out.println("evenIndicies produces: \t"+nums_ind);
47          System.out.println("evenValues produces: \t"+nums_val);
48          System.out.println("reverse produces: \t"+nums_rev);
49          System.out.println("last one produces: \t"+nums_lwbf);
```

```
50
51     }
52
53 }
```

▼ **_MACOSX/.ArrayTest.java**

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